TI TIDA-060029 analog front end

Matched RG/RF: 100R / 5k / 100k

File: afe.kicad_sch

Codec DAC excitation

Coherent codec DAC excitation

File: source.kicad_sch

Codec acquisition (TAC5112)

Two channels / one frame clock

File: codec.kicad_sch

Range relays

Relay-selected ranges

File: range.kicad_sch

Jack protection (Rev G)

Jack protection and isolation

File: protect.kicad_sch

USB, RP2040 and boot / debug

USB control, firmware and host calibration

File: digital.kicad_sch

Power: 5V input, forced PWM, quiet analog rails

External 5V → quiet +/-12V

File: power.kicad_sch

Engineering prototype. One PCB. Isolated, discharged DUT only.

Temper

Sheet: /
File: temper-lcr.kicad_sch

Title: Temper LCR — TIDA-060029 based

Size: A3	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 8/8

U6A source buffer; U6B HI sense; U7A transimpedance; U7B LO sense.
Ranges: latching relays K1/K2/K3 (range sheet) close one matched RG/RF pair.
Four-terminal BNC jacks: HCUR = DUT_HI, LCUR = DUT_LO; followers sense HPOT / LPOT.
One local C104=100pF compensates all ranges; C78 100pF DUT_LO to GND (Rev E stability).
No BUF634A: maximum source current approximately 18mA. Isolated, discharged DUT only.

TI source and range network

U6A
OPA2810IDGKR
EXCITE
SRC_DRV
R104
47R
SOURCE_BUF
Rev E: R104 47R isolates the SOURCE_BUF relay bus from U6A.

R40
RG100
100R 0.1%
R41
RG5K
5k 0.1%
R42
RG100K
100k 0.1%
DUT_HI

J3
JK_HCUR
GND
BNC HCUR
J7
JK_HPOT
BNC HPOT
J8
JK_LPOT
BNC LPOT
J9
JK_LCUR
BNC LCUR
J3 HCUR, J7 HPOT, J8 LPOT, J9 LCUR: BNC, 22 mm pitch, shells to GND.
Connect an external, isolated, discharged DUT with Kelvin leads.
R105/R106 1k tie each POT to its CUR for two-terminal leads.

TI transimpedance topology; Rev C compensation
C104=100pF directly across U7.1 and U7.2 in all ranges

Rev D: R94 22R 0.1% between LCUR and the TIA summing node.
Rev E: LOW follower senses LPOT; host uses Zf = R94 + RF||C104.

R94
DUT_LO
22R 0.1%
TIA_SUM
C78
100pF C0G 5%
Rev E: C78 100pF C0G from LCUR (DUT_LO) to GND (TIA margin with relay parasitics).

R43
RF100
100R 0.1%
R44
RF5K
5k 0.1%
R45
RF100K
100k 0.1%
C104
100pF C0G 5%
U7A
GND
TIA_OUT
OPA2810IDGKR

U6B
HPOT
HIGH_BUF
OPA2810IDGKR

U7B
LPOT
LOW_BUF
OPA2810IDGKR

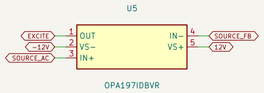
Independent HI / LO voltage followers

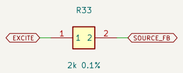
U6C
U7C
OPA2810IDGKR
12V
-12V
C42
100nF 50V
C43
100nF 50V
C44
1uF 50V
C45
1uF 50V
C50
10nF 50V
C51
10nF 50V
C46
100nF 50V
C47
100nF 50V
C48
1uF 50V
C49
1uF 50V
C52
10nF 50V
C53
10nF 50V

Retain TI 100nF + 1uF bypass per rail; added 10nF at each supply pin.
PCB plane windows and short feedback routes are specified in reference-conformance.md.

Temper
Sheet: /TI TIDA-060029 analog front end/
File: afe.kicad_sch
Title: TI TIDA-060029 analog front end
Size: A2 | Date: 2026-10-20 | Rev: E
KiCad E.D.A. 10.0.4 | Id: 2/8

TAC5112 OUT1P/OUT1M differential line out at fs = 187.5kHz.
OPA810 difference stage: 2.2k in, 2.0k || 280pF (2 x 560pF in series) feedback and ground leg, gain 0.909.
DC coupled: the stage rejects the 1.65V DAC common mode.



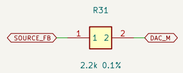




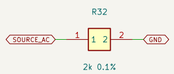




















Global net names connect functional sheets. See TI conformance matrix.

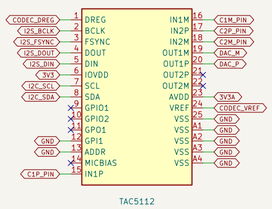
Temper

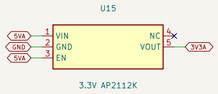
Sheet: /Codec DAC excitation/
File: source.kicad_sch

Title: Codec DAC excitation

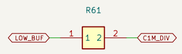
Size: A2	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 3/8

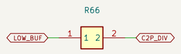
CH1 = HIGH_BUF − LOW_BUF (IN1P/IN1M); CH2 = LOW_BUF − TIA_OUT (IN2P/IN2M).
Each leg: 1k series, 1k + 560pF to GND (K = 0.5), 1uF, BAT54S clamp, 470R to the pin.
TDM controller: RP2040 drives BCLK 12MHz / FSYNC 187.5kHz. I2C address 0x50.
AVDD from U15 AP2112K−3.3 on 5VA; IOVDD from 3V3.

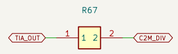




















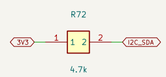


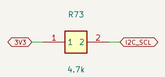


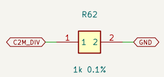


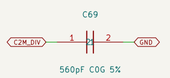






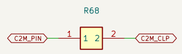


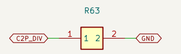








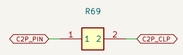


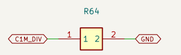






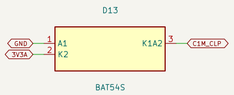


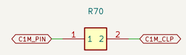


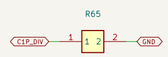








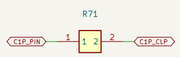












Global net names connect functional sheets. See TI conformance matrix.

Temper

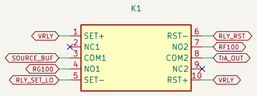
Sheet: /Codec acquisition (TAC5112)/
File: codec.kicad_sch

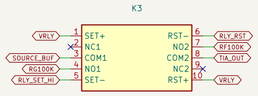
Title: Codec acquisition (TAC5112)

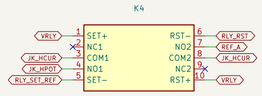
Size: A2	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 4/8

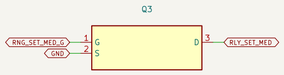
K1 100R, K2 5k, K3 100k, K4 reference: Panasonic TQ2SA-L2-5V dual-coil latching DPDT.
Pole 1: SOURCE_BUF -> RGx. Pole 2: TIA_OUT -> RFx. NC contacts unconnected.
RESET coils (K1-K4) in parallel on Q2; SET coils on Q3/Q4/Q5/Q6. 1N4148WS flyback per drain.
K4 reference: COM2 on HCUR, NO2 through R107 1k 0.1% to LCUR (REFCAPTURE, leads unplugged).
VRLY = 5V through R101 2.2R + C112 22uF. Firmware: source off, RESET 20ms, SET 20ms.



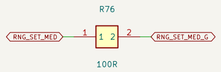


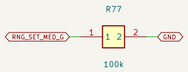




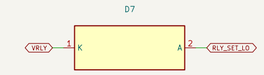


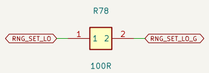


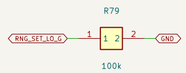




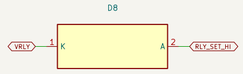


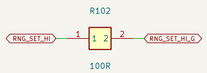


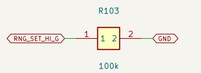




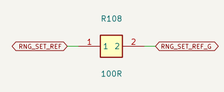


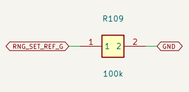


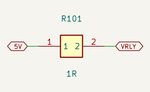














Global net names connect functional sheets. See TI conformance matrix.

Temper

Sheet: /Range relays/
File: range.kicad_sch

Title: Range relays

Size: A2	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 5/8

Each BNC: H4V5HD3B TVS to GND at the centre pin, then a normally-open TLP172AM photoMOS.
Series 10R (HCUR / LCUR, CRCW0805-HP) and 100R (HPOT / LPOT, CRCW0603-HP) after the switch;
second stage H3V3LD5B at DUT_HI / HPOT / LPOT (+ 1nF C0G on the POTs), BAV99 pair at DUT_LO.
R105 / R106 and K4 (both poles) on the jack side. LED strings from VRLY, AO3400A on GPIO15.
Residual-voltage senses: 1.5M from HCUR / LCUR to GPIO28 / 29, 240k to 3V3, 240k + 100nF to GND.

R105
JK_HCUR
1
1 2
2
JK_HPOT
1k

R106
JK_LCUR
1
1 2
2
JK_LPOT
1k

D21
JK_HCUR
1
A1
A2
2
GND
H4V5HD3B

U18
ISO_S1M
ISO_SINK
1
3
A
K
D
D
4
6
JK_HCUR
ISO_HCUR
TLP172AM

R122
ISO_HCUR
1
1 2
2
DUT_HI
10R

D22
JK_HPOT
1
A1
A2
2
GND
H4V5HD3B

U19
ISO_S1
ISO_S1M
1
3
A
K
D
D
4
6
ISO_HPOT
JK_HPOT
TLP172AM

R123
ISO_HPOT
1
1 2
2
HPOT
100R

D23
JK_LPOT
1
A1
A2
2
GND
H4V5HD3B

U20
ISO_S2
ISO_S2M
1
3
A
K
D
D
4
6
ISO_LPOT
JK_LPOT
TLP172AM

R124
ISO_LPOT
1
1 2
2
LPOT
100R

D24
JK_LCUR
1
A1
A2
2
GND
H4V5HD3B

U21
ISO_S2M
ISO_SINK
1
3
A
K
D
D
4
6
JK_LCUR
ISO_LCUR
TLP172AM

R125
ISO_LCUR
1
1 2
2
DUT_LO
10R

D25
DUT_HI
1
A1
A2
2
GND
H3V3LD5B

D26
HPOT
1
A1
A2
2
GND
H3V3LD5B

D27
LPOT
1
A1
A2
2
GND
H3V3LD5B

C119
HPOT
1
1 2
2
GND
1nF C0G

C120
LPOT
1
1 2
2
GND
1nF C0G

D28
GND
GND
1
2
A1
K2
K1A2
3
DUT_LO
BAV99

R126
JK_HCUR
1
1 2
2
JACK_HC
1.5M

R127
JACK_HC
1
1 2
2
3V3
240k

R128
JACK_HC
1
1 2
2
GND
240k

C121
JACK_HC
1
1 2
2
GND
100nF 50V

R129
JK_LCUR
1
1 2
2
JACK_LC
1.5M

R130
JACK_LC
1
1 2
2
3V3
240k

R131
JACK_LC
1
1 2
2
GND
240k

C122
JACK_LC
1
1 2
2
GND
100nF 50V

R132
VRLY
1
1 2
2
ISO_S1
300R

R133
VRLY
1
1 2
2
ISO_S2
300R

Q7
ISO_G
GND
1
2
G
S
D
3
ISO_SINK
AO3400A

R134
ISO_DRIVE
1
1 2
2
ISO_G
100R

R135
ISO_G
1
1 2
2
GND
100k

Global net names connect functional sheets. See TI conformance matrix.

Temper

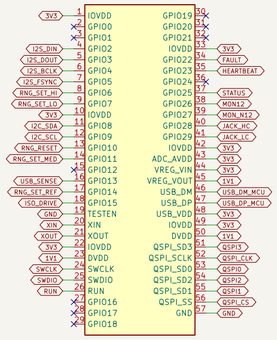
Sheet: /Jack protection (Rev G)/
File: protect.kicad_sch

Title: Jack protection (Rev G)

Size: A2	Date: 2026-10-20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 6/8

External 5V input powers the instrument. USB VBUS is sensed only; no backfeed.
Native USB CDC; 120MHz MCU; dual-lane ADC acquisition at up to 1MSPS.
Hold BOOTSEL while resetting for ROM USB firmware loading. SWD and UART provided.

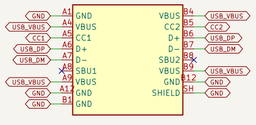
U1



U2



J1



D1



Y1



SW1



SW2



D4



J4



J5



R1



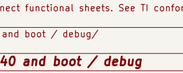
R2

R3

R4

R5

R6



R7

R8

R9

R10

R12

R13

C1

C2

C3

C4

C5

C6

C7

C8

C9

C10

C11

C12

C13

C14

C15

R98

R99

C110

C111

R113

D15

R114

D16

R115

D17

R116

D18

R117

D19

R118

D20

Global net names connect functional sheets. See TI conformance matrix.

Temper	
Sheet: /USB, RP2040 and boot / debug/	
File: digital.kicad_sch	
Title: USB, RP2040 and boot / debug	
Size: A2	Date: 2026-10-20
KiCad E.D.A. 10.0.4	**Rev: E**
	Id: 7/8

Input: REGULATED 5V +/−5%, 1A, center positive. USB does not supply this board.
TPS65131 PSP/PSN grounded: forced PWM 1.25−1.5MHz. RAW_P −14.31V: RAW_N −−14.39V.
TPS7A49 −+11.97V: TPS7A30 −−12.00V. Fixed output values are nominal.
Do not substitute capacitor voltage/package without checking DC−bias derating.

U3
3.3V AP2112K

U11
TPS7A4901DGNR

U12
TPS7A3001DGNR

U13
TPS65131RGER

U16
TPS2553

D2
SS14

D3
SS14

L1
4.7uH 3.8A shielded

L2
4.7uH 3.8A shielded

FB1
100R@100MHz 800mA

R95
1R

R96
1R

R80
100R

R81
430k

R82
40.2k

R83
240k

R84
150k

R85
82k

R86
9.1k

R87
100k 0.1%

R88
1k 0.1%

R89
5k 0.1%

C16
1uF 50V

C17
1uF 50V

C18
10uF 25V X5R

C80
10uF 50V X7R

C81
100nF 50V

C82
10uF 50V X7R

C83
100nF 50V

C84
220nF 25V

C85
4.7nF 50V

C86
10nF 50V

C87
15pF C0G

C88
4.7pF C0G

C89
10uF 50V X7R

C90
10uF 50V X7R

C91
10uF 50V X7R

C92
10uF 50V X7R

C93
10nF 50V

C94
10uF 50V X7R

C95
10uF 50V X7R

C96
10nF 50V

C97
100nF 50V

C98
10nF 50V

C99
10uF 50V X7R

C100
10uF 50V X7R

C101
10nF 50V

C102
100nF 50V

C103
22uF 25V X5R

R97
100k

R100
150k

R110
24k 1%

C113
4.7uF 25V X5R

R112
5k 0.1%

R111
1.5M

R119
100k

R120
100k

C116
100nF 50V

C118
100nF 50V

R121
82k

C117
1uF 50V

PWR_FLAG

Global net names connect functional sheets. See TI conformance matrix.

Temper		
Sheet: /Power: 5V input, forced PWM, quiet analog rails/		
File: power.kicad_sch		
Title: Power: 5V input, forced PWM, quiet analog rails		
Size: A2	Date: 2026−10−20	**Rev: E**
KiCad E.D.A. 10.0.4		Id: 8/8